



COMP3055

Machine Learning

Topic 5 – Instance Based Learning

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2025 Autumn

Motivation

Classify whether a customer will respond to a survey question using **Instance Based Learning**



Instance Based Learning

- Directly compare new problem instances with instances seen in training
直接比较：输入新的样本时，算法计算该样本与训练集中样本的相似度；
- No explicit modeling of the training data
无模型假设：算法不需要训练参数，预测完全依赖已有样本；
- Complexity grows with the training data
计算复杂度高：每次预测都要扫描整个训练集；
- Classical instance based learning technique
 - **K Nearest Neighbour**
典型代表算法：K 最近邻 (K-Nearest Neighbour, KNN)。

K-Nearest Neighbour Model

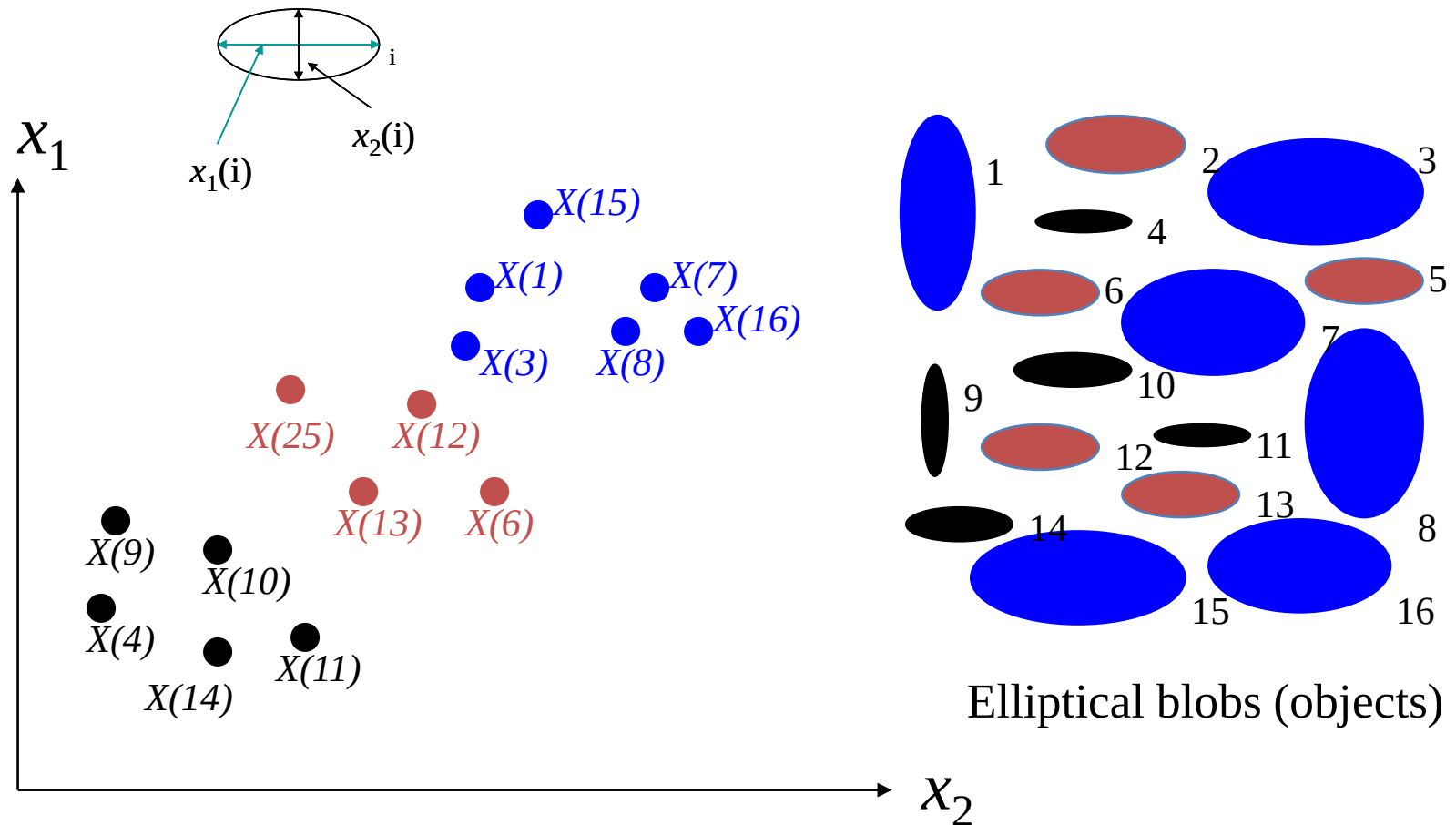
Example

Classify whether a customer will respond to a survey question using a 3-Nearest Neighbour classifier.

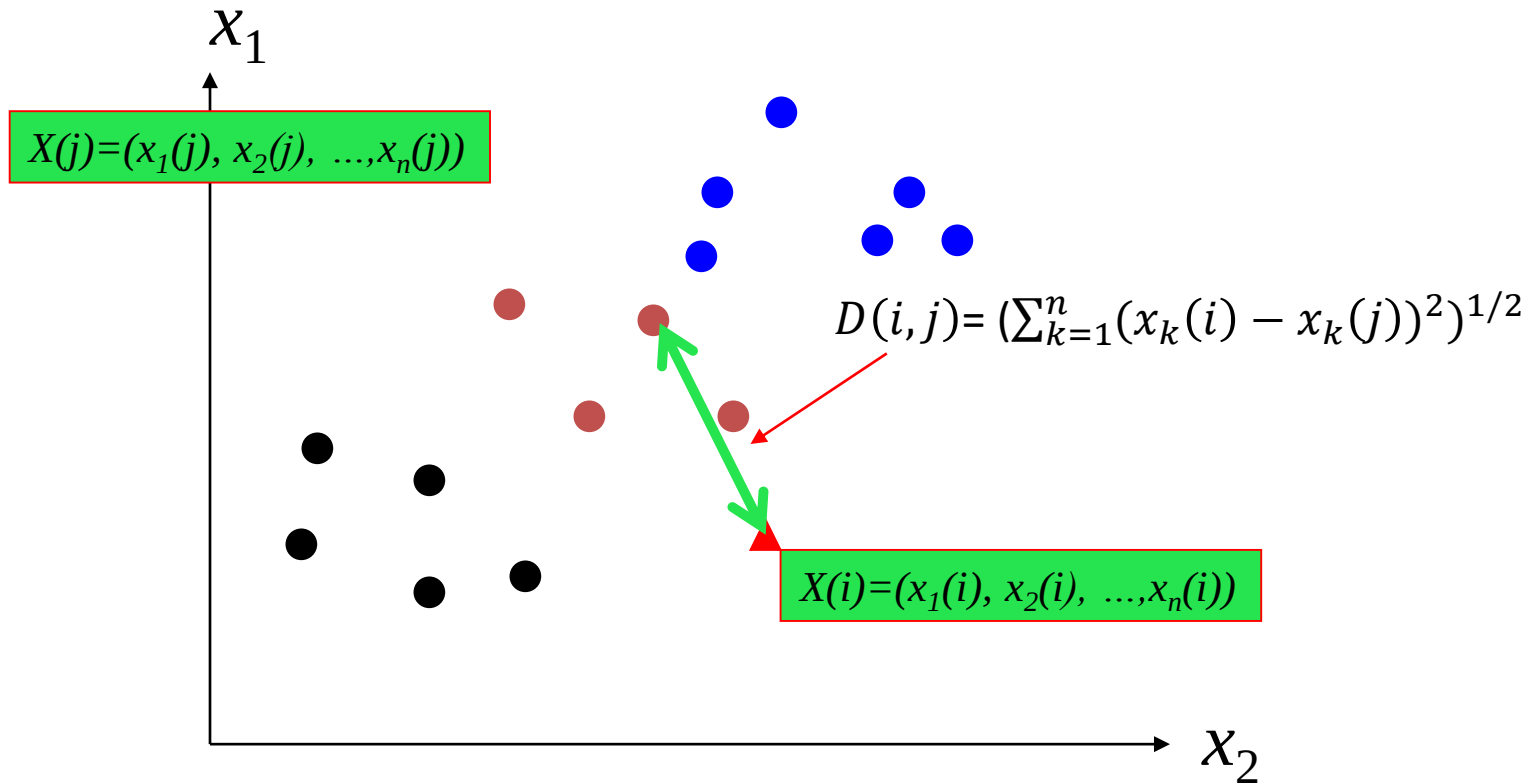


Customer	Age	Income	No. credit cards	Response
John	35	35K	3	No
Rachel	22	50K	2	Yes
Hannah	63	200K	1	No
Tom	59	170K	1	No
Nellie	25	40K	4	Yes
David	37	50K	2	?

Objects, Feature Vectors, Points



Nearest Neighbours



Nearest Neighbour Algorithm

Given training data $(X(1), D(1)), (X(2), D(2)), \dots, (X(N), D(N))$,

Define a distance metric between points in inputs space.

Common measures are:

Euclidean Distance

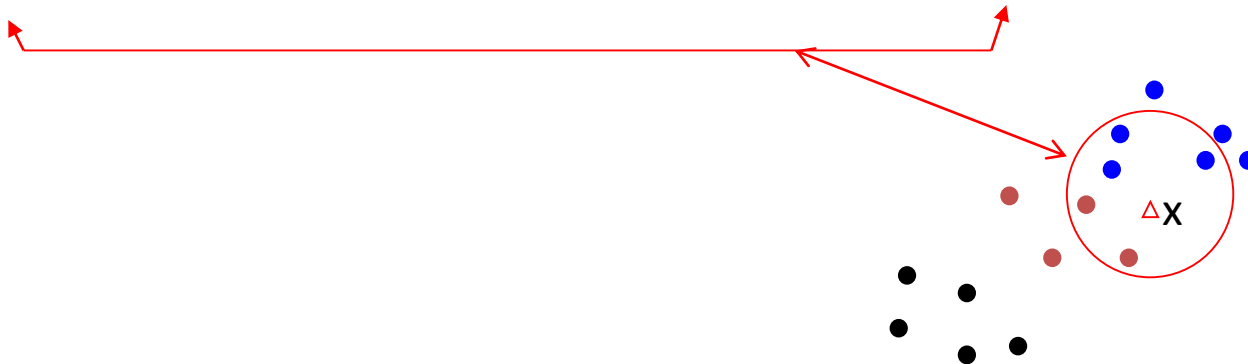
$$D(i, j) = (\sum_{k=1}^n (x_k(i) - x_k(j))^2)^{1/2}$$

K-Nearest Neighbour Model

Given test point X

- Find the K nearest training inputs to X
- Denote these points as

$(X(1), D(1)), (X(2), D(2)), \dots, (X(k), D(k))$



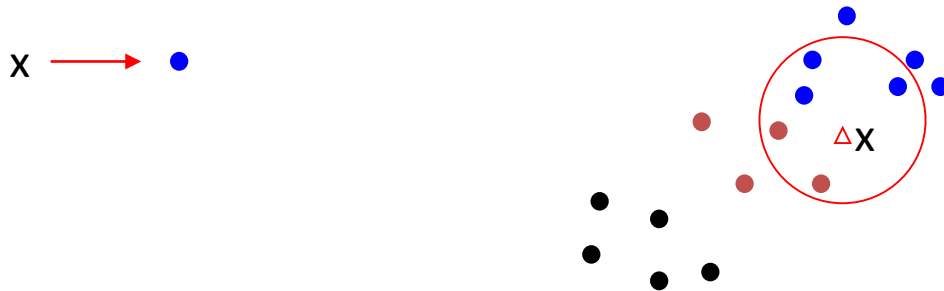
K-Nearest Neighbour Model

Instance based learning

The class identification of X

$Y = \text{most common class in set } \{D(1), D(2), \dots, D(k)\}$

Majority rule 多数投票原则



K-Nearest Neighbour Model

Example

Classify whether a customer will respond to a survey question using a 3-Nearest Neighbour classifier.

Customer	Age	Income	No. credit cards	Response
John	35	35K	3	No
Rachel	22	50K	2	Yes
Hannah	63	200K	1	No
Tom	59	170K	1	No
Nellie	25	40K	4	Yes
David	37	50K	2	?

K-Nearest Neighbour Model

Example

3-Nearest Neighbours

- Euclidean Distance Calculation
 - **Data Preprocessing on Income**
 - By removing 'k', e.g., 37k $\rightarrow$ 37, 35k $\rightarrow$ 35
 - **Define Feature Vector**
 - Feature Vector David = (37, 50, 2)
 - Feature Vector John = (35, 35, 3)
 - **Calculate Euclidean Distance**
 - Euclidean Distance (David, John) = $\sqrt{(37 - 35)^2 + (50 - 35)^2 + (2 - 3)^2}$
$$= \sqrt{2^2 + 15^2 + (-1)^2}$$
$$= \sqrt{4 + 225 + 1}$$
$$= \sqrt{230} \approx 15.17$$

K-Nearest Neighbour Model

Example

3-Nearest Neighbours

Customer	Age	Income	No. credit cards	Response
John	35	35K	3	No
Rachel	22	50K	2	Yes
Hannah	63	200K	1	No
Tom	59	170K	1	No
Nellie	25	40K	4	Yes
David	37	50K	2	?

The diagram illustrates the distances from David to his 3 nearest neighbors. The distances are labeled as follows:

- Distance to Nellie: 15.74
- Distance to Tom: 122
- Distance to Hannah: 152.23
- Distance to Rachel: 15
- Distance to John: 15.16

K-Nearest Neighbour Model

Example

3-Nearest Neighbours

Customer	Age	Income	No. credit cards	Response
John				No
Rachel				Yes
Hannah	63	200K	1	No
Tom	59	170K	1	No
Nellie				Yes
David	37	50K	2	?

Distances from David to nearest neighbors:

- Hannah: 15.74
- Tom: 122
- Nellie: 152.23
- Rachel: 15
- John: 15.16

Three nearest ones to David are: No, Yes, Yes

David 预测为 “Yes”

K-Nearest Neighbour Model

Example

3-Nearest Neighbors

Customer	Age	Income	No. credit cards	Response
John				No
Rachel				Yes
Hannah	63	200K	1	No
Tom	59	170K	1	No
Nellie				Yes
David	37	50K	2	Yes

Distances from David to other customers:

- David to Nellie: 15.74
- David to Tom: 122
- David to Hannah: 152.23
- David to Rachel: 15
- David to John: 15.16

Three nearest ones to David are: No, Yes, Yes

K-Nearest Neighbour Model

Picking K

- Use *N fold cross validation* – Pick K to minimize the cross validation error 使用 N 折交叉验证 (N-fold cross validation) 来确定最佳的 K。
 - 1 对每个训练样本：
 - 找出它的 K 个最近邻；
 - 基于这些邻居进行分类；
 - 计算分类错误率。
 - 2 对所有样本取平均错误率。
 - 3 选择使平均错误率最小的 K 值作为最优 K。
 - For each of N training example
 - Find its K nearest neighbours
 - Make a classification based on these K neighbours
 - Calculate classification error
 - Output average error over all examples
 - Use the K that gives lowest average error over the N training examples
-  结论：
- K 太小 → 模型过拟合 (对噪声敏感)；
 - K 太大 → 模型欠拟合 (边界模糊)。

K-Nearest Neighbour Model

Distance Metrics

- Euclidean Distance

$$d(x, y) = \sqrt{\sum_i (x_i - y_i)^2}$$

- Manhattan Distance

$$d(x, y) = \sum_i |x_i - y_i|$$

- Hamming distance

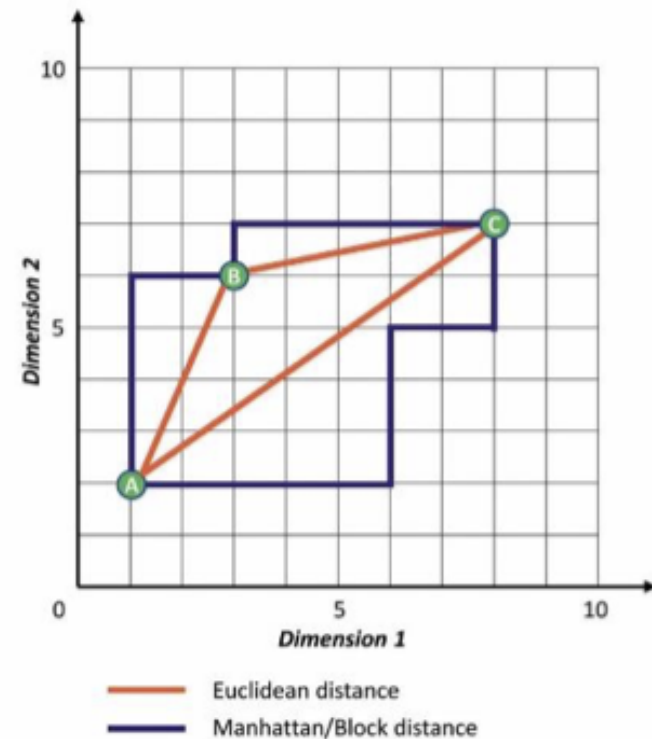
$$d(x, y) = \frac{1}{N} \sum_i 1_{x_i \neq y_i}$$

K-Nearest Neighbour Model

Distance Metrics

- Manhattan Distance

$$d(x, y) = \sum_i |x_i - y_i|$$



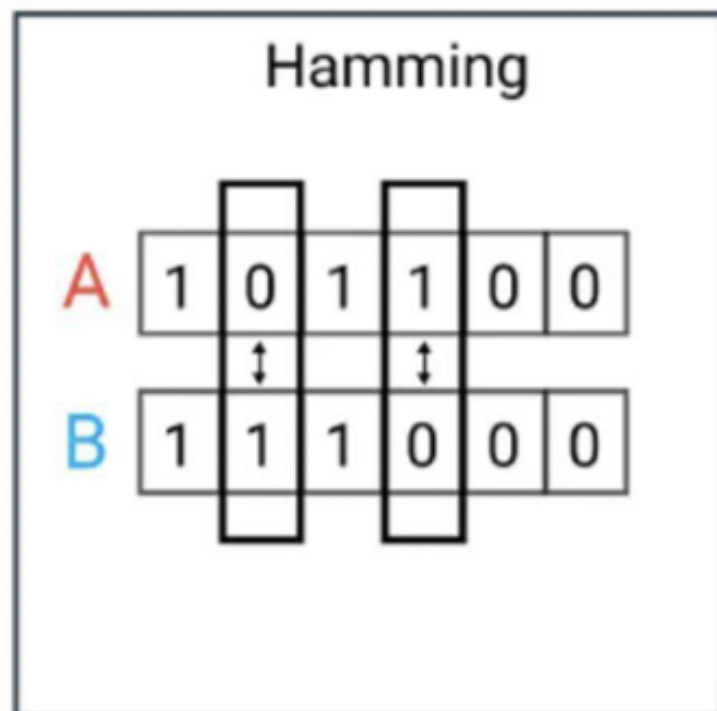
distance	block	euclidean	difference
$\overline{AB}$	6	$\sqrt{4^2 + 2^2} = 4,47$	1,53
$\overline{BC}$	6	$\sqrt{5^2 + 1^2} = 5,10$	0,90
$\overline{AC}$	12	$\sqrt{7^2 + 5^2} = 8,60$	3,40
sum	24	18,17	5,83

K-Nearest Neighbour Model

Distance Metrics

- Hamming Distance

$$d(x, y) = \frac{1}{N} \sum_i 1_{x_i \neq y_i}$$



K-Nearest Neighbour Model

Q&A

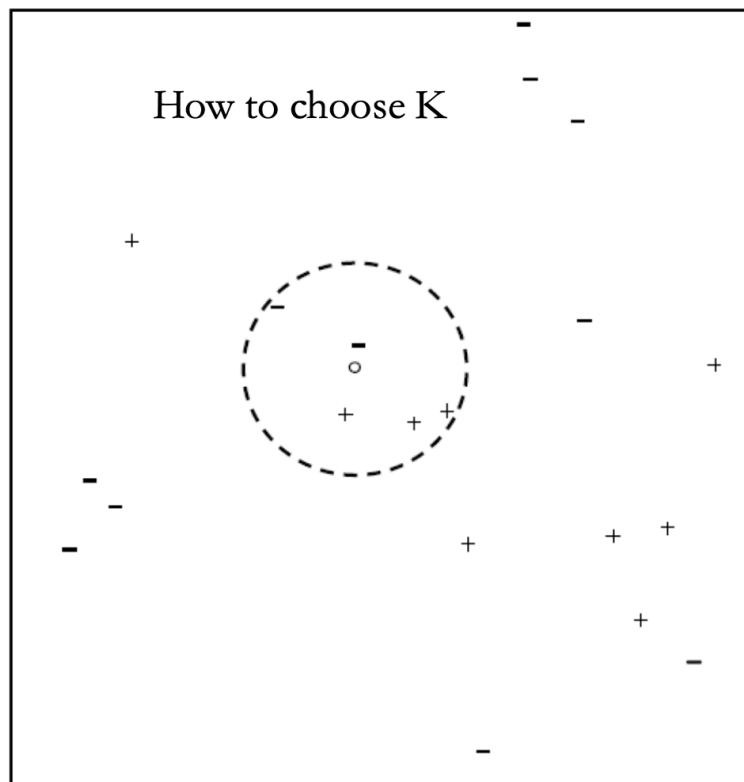
For the example we saw earlier, pick the best K from the set {1, 2, 3} to build a K-NN classifier.

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K-Nearest Neighbour Model

Q&A

For the example we saw earlier, pick the best K from the set {1, 2, 3} to build a K-NN classifier.



- Loop through and see what the evaluation measure is for different values of k
- More details in next week lab

K-Nearest Neighbour Model

Q&A

For the example we saw earlier, pick the best K from the set {1, 2, 3} to build a K-NN classifier.

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Distances from David to other customers:

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K-Nearest Neighbour Model

Pros and Cons

• Pros

- Simple, effective; no training required; optimal with infinite data

• Cons

- Have to select k ; issues with unbalanced classes; often slow (need to find those k -neighbours); features must be selected carefully

1 简单直观

- 无需建立复杂模型;
- 思想直白: 相似的点 $\rightarrow$ 相似的标签。

2 无训练阶段

- 属于“懒惰学习” (lazy learning);
- 训练过程几乎为零, 只需保存数据。

3 理论上最优 (当数据无限时)

- 在无限数据条件下, KNN 可逼近最优贝叶斯分类器 (Bayes Optimal)。

1 K 值选择困难

- 过小 $\rightarrow$ 对噪声敏感;
- 过大 $\rightarrow$ 模型平滑、失去细节。

2 不平衡数据问题

- 若某类样本数远多于另一类, 容易导致偏向多数类。

3 计算复杂度高

- 每次预测都要计算所有样本的距离;
- 随着数据量增加, 预测会变得很慢。

4 特征选择敏感

- 不同量纲的特征会影响结果;
- 通常需要归一化 (normalization)。

Further Readings

Chapter 8, T. M. Mitchell, Machine Learning,
McGraw-Hill International Edition, 1997

1 欧几里得距离计算公式必须熟记：

$$D(i, j) = \sqrt{\sum (x_i - x_j)^2}$$

2 多数投票法 (Majority Rule) 原理要清楚。

3 过拟合与欠拟合在 K 的变化趋势：

- 小 K → 过拟合

- 大 K → 欠拟合

4 K 的选择常通过 交叉验证 实现。

5 特征必须进行 归一化 (Normalization)，否则“量纲大的特征”会主导结果。

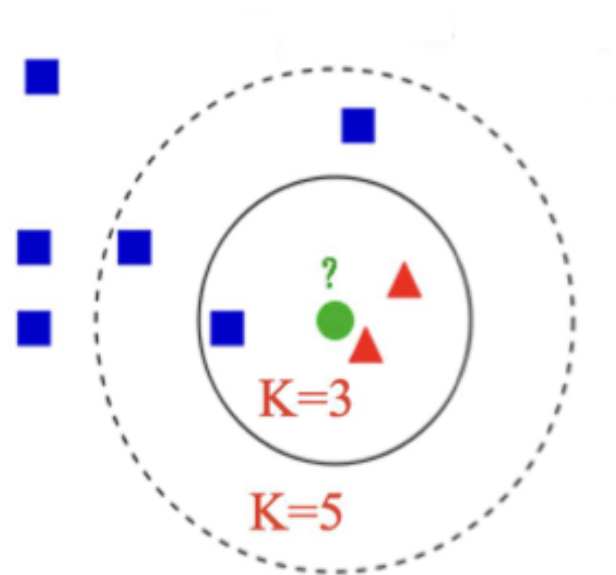
Any Questions?



Question

For a green point, what will 3NN predict? What will 5NN predict? Setting K to a large value seems like a good idea. We get more votes! Given this particular training set, would you recommend setting $K = 8$?

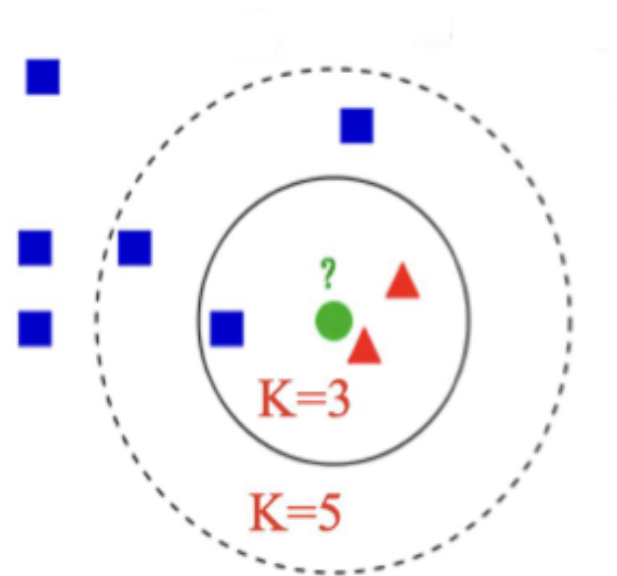
- A) Red, Blue, Yes**
- B) Red, Blue, No**
- C) Red, Blue, Cannot decide**
- D) Blue, Blue, Cannot decide**



Answer

For a green point, what will 3NN predict? What will 5NN predict? Setting K to a large value seems like a good idea. We get more votes! Given this particular training set, would you recommend setting $K = 8$?

- A) Red, Blue, Yes
- B) Red, Blue, No**
- C) Red, Blue, Cannot decide
- D) Blue, Blue, Cannot decide



Answer

For a green point, what will 3NN predict? What will 5NN predict? Setting K to a large value seems like a good idea. We get more votes! Given this particular training set, would you recommend setting $K = 8$?

There are only 2 red instances in the training set. Therefore, any value of $K \geq 5$ would have a majority of blue instances.

